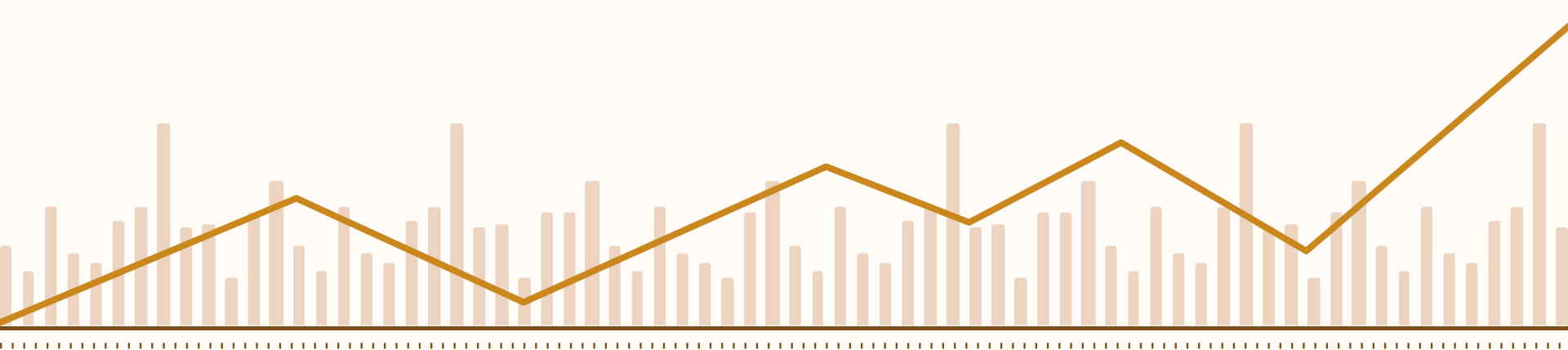




Headline Sentiment Stock Predictions

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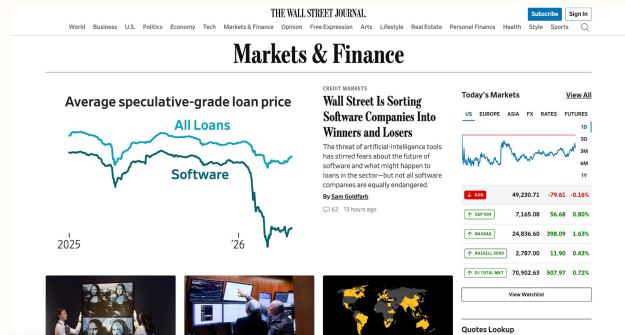
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Introduction: Headline-Based Predictions

- Goal: Use top 25 headlines on a given day to predict whether the Dow Jones Industrial Average (DIJA) will increase or decrease
 - Index of 30 large stocks
- Data includes inputs of date, top 25 headlines and label
- Use various feature selection methods and model types and review performance



Motivation

Our motivation

- Collective group interest in finance and stock prediction
- Combining stock market interest with NLP skills

Other Interest

- Potential of alpha generation
- See relationship between headlines, investors and prices

Learning Hope

- Observe whether or not headlines can be predictors of market movement
- What features/models predict price movement best

Our Goal

- Show how different text features can be used to predict markets
- Outperform baselines with our models

Data: Overview

- Dataset from Kaggle
- 25 news headlines per day from Reddit WorldNews Channel
- 1989 days of data (8/6/08-7/1/16)
- Contained labels for each date – 1 for increase, 0 otherwise
- Split data into 1600 training samples and 389 testing samples
 - Chosen randomly
- Libraries Used: Numpy, Pandas, Datasets

The Kaggle logo, featuring the word "kaggle" in a blue, lowercase, sans-serif font, is positioned in the lower right area of the slide. It is set against a light blue rectangular background.

Data: Example

Date	Label	Top1	Top2	...	Top25
2010-09-21	1	Germany is planning a direct rail link between Frankfurt and London: Journey time under 4 hours.	Oktoberfest 2010 - The Big Picture [pics]		UK Proposes All Paychecks Go to the State First - Everybody should just work at their job and the government should just give everyone what allowance they feel they deserve to get.

Data: Visualization (All Days)



Data: Visualization (TF-IDF Regulated)

DJIA Up Days (Label = 1)



DJIA Down Days (Label = 0)



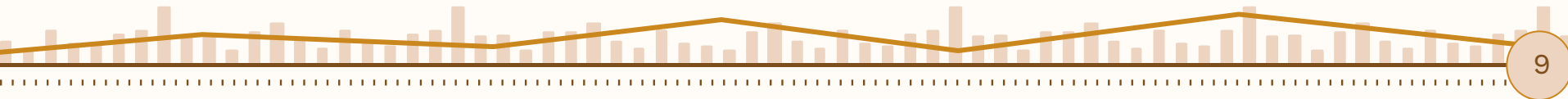
Data: Problems and Findings

Problems:

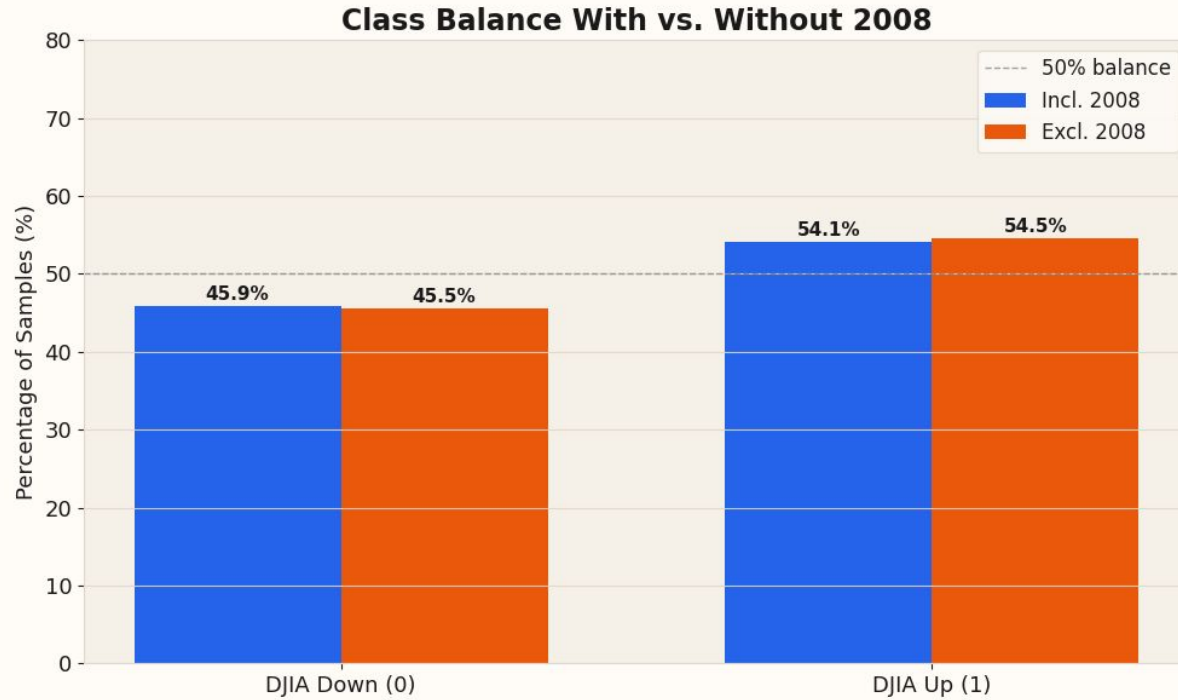
- Meaning of labels were unclear in the dataset
 - Confirm the meaning of the binary labels in relation to DJIA behavior
- Test the accuracy of the labels
- 2008 financial crisis included in dataset

Findings:

- (1) → DJIA increased
- (0) → DJIA decreased
- A handful of labels were incorrect



Data: Balance With vs Without 2008



Methods: Overview

Project Goal

Use headline data to predict price movement of the Dow Jones

Features

Bag-of-words, Word2Vec, VADER (sentiment analysis tool)

Classifiers

Logistic Regression, KNN, Naive Bayes, CNN

- Used Scikit and TensorFlow

LLMs

Additional testing with BERT, Mistral

Methods: Baselines

Baselines Used:

- Majority Baseline: Assign all labels the majority label (1)
- Random Baseline: Assign all of the labels randomly
- Momentum Baseline: Assign label based on the price movement of previous day

Chosen Baselines	Baseline Accuracy
Random Baseline	0.4987
Majority Baseline	0.4833
Previous Day Momentum Baseline	0.4807

Features: Bag Of Words

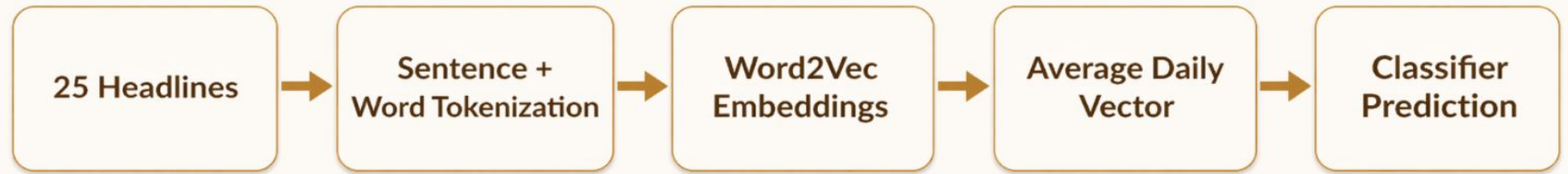
- Used NLTK stopwords
- Broke each headline to words, added words to positive/negative feature list
- Extracted the 1000 most common words
 - Served as features for model input

Most Frequent Words by Price Label

Word #	Positive	Negative
1	us	us
2	government	police
3	police	government
4	israel	world
5	china	israel
6	war	war
7	world	china
8	years	years
9	one	killed
10	president	one

Features: Word2Vec

- Combined 25 headlines into one daily text field
- Tokenized with sent_tokenize + word_tokenize
- Trained Word2Vec on training headlines only
- Converted each day to one vector by averaging word embeddings



Features: VADER (Sentiment Tool)

August 11, 2008: DJIA - (1)

- Daily headlines are assigned a score ranging from -1 (most negatively associated), to 1 (most positively associated)
- Each group of daily headlines is summarized into 8 descriptors per day
 - Used as 8-feature vector inputs when fed to models

Feature	Value
Avg. Sentiment Score	-0.1769
Sentiment Spread (Std. Dev.)	+0.3696
Most Negative Headline	-0.8271
Most Positive Headline	+0.7783
Avg. Positivity	+0.0711
Avg. Negativity	+0.1896
Share of Positive Headlines	12.0%
Share of Negative Headlines	64.0%

Methods: BERT



Model Configuration

- Used **distilbert-base-uncased** (40% smaller, 60% faster)
- WordPiece tokenizer with truncation
- Fine-tuned on labeled training data

Training Hyperparameters

- **Duration:** 5 epochs
- **Learning Rate:** $2e-5$
- **Weight Decay:** 0.01

Methods: Mistral

Mistral Implementation

- Model: **mistral-large-latest**
- Tested **three strategies**: zero, one, and three shot
- Shot selection
 - All from different years
 - Two examples with label 1, one with label 0

Prompt Engineering

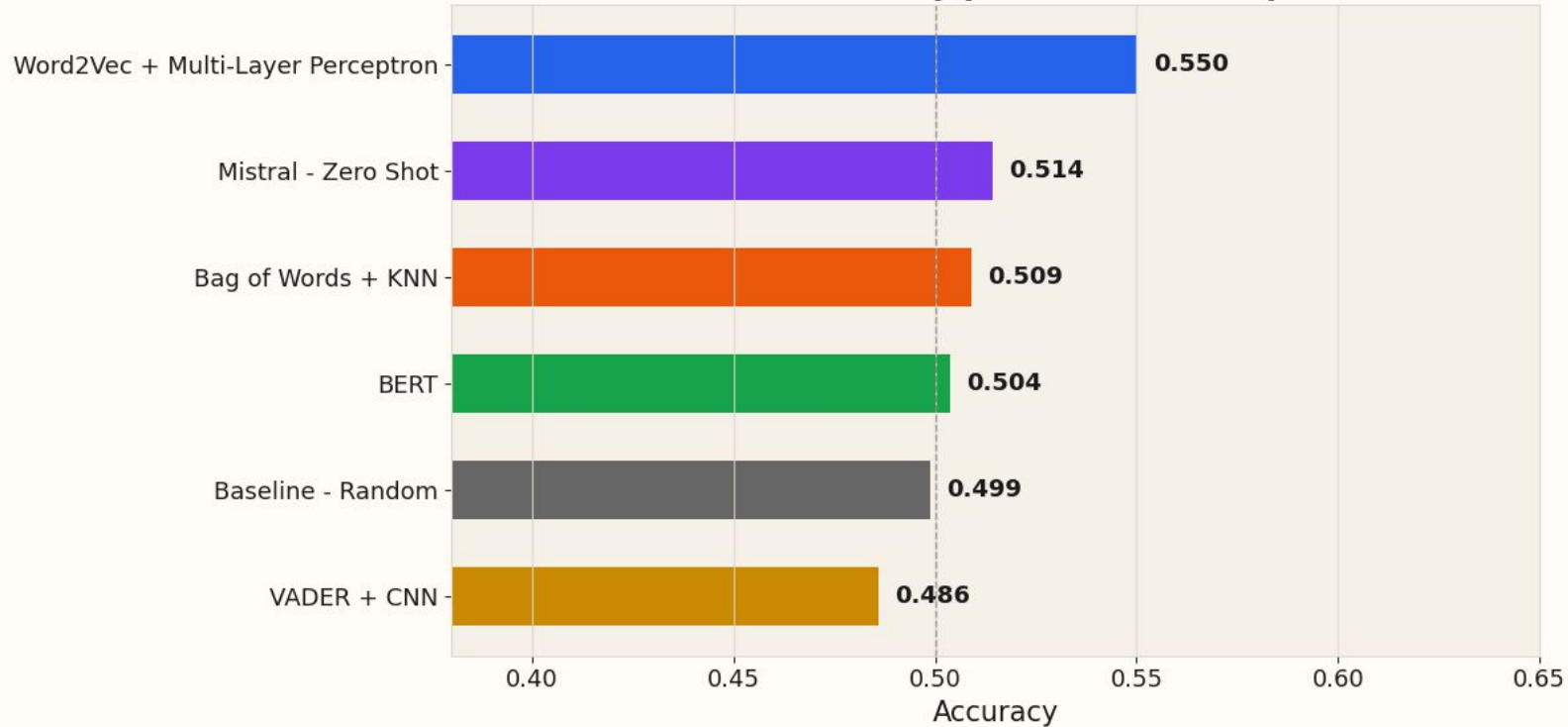
- Role: **Financial Analyst**
- Initial test: Plain prompt comparison
- **Structure:**
`role → task → examples → headlines → "Answer:"`

Example Prompt Excerpt

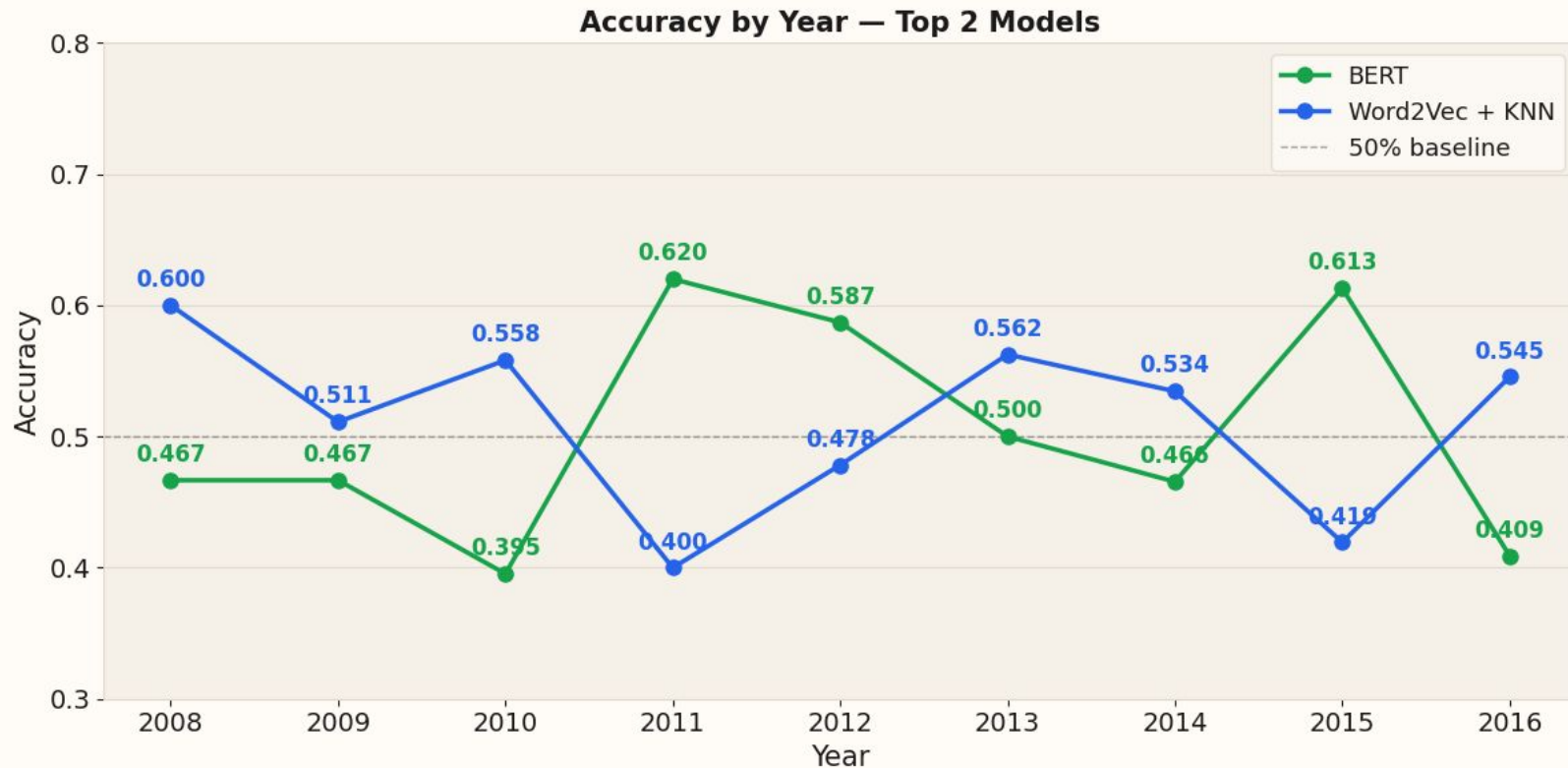
"You are a financial analyst. Based on the following 25 news headlines... predict whether the Dow Jones Industrial Average (DJIA) closed higher or lower... Respond with only '1' or '0'."

Results: Accuracy

Best Accuracy per Feature Group

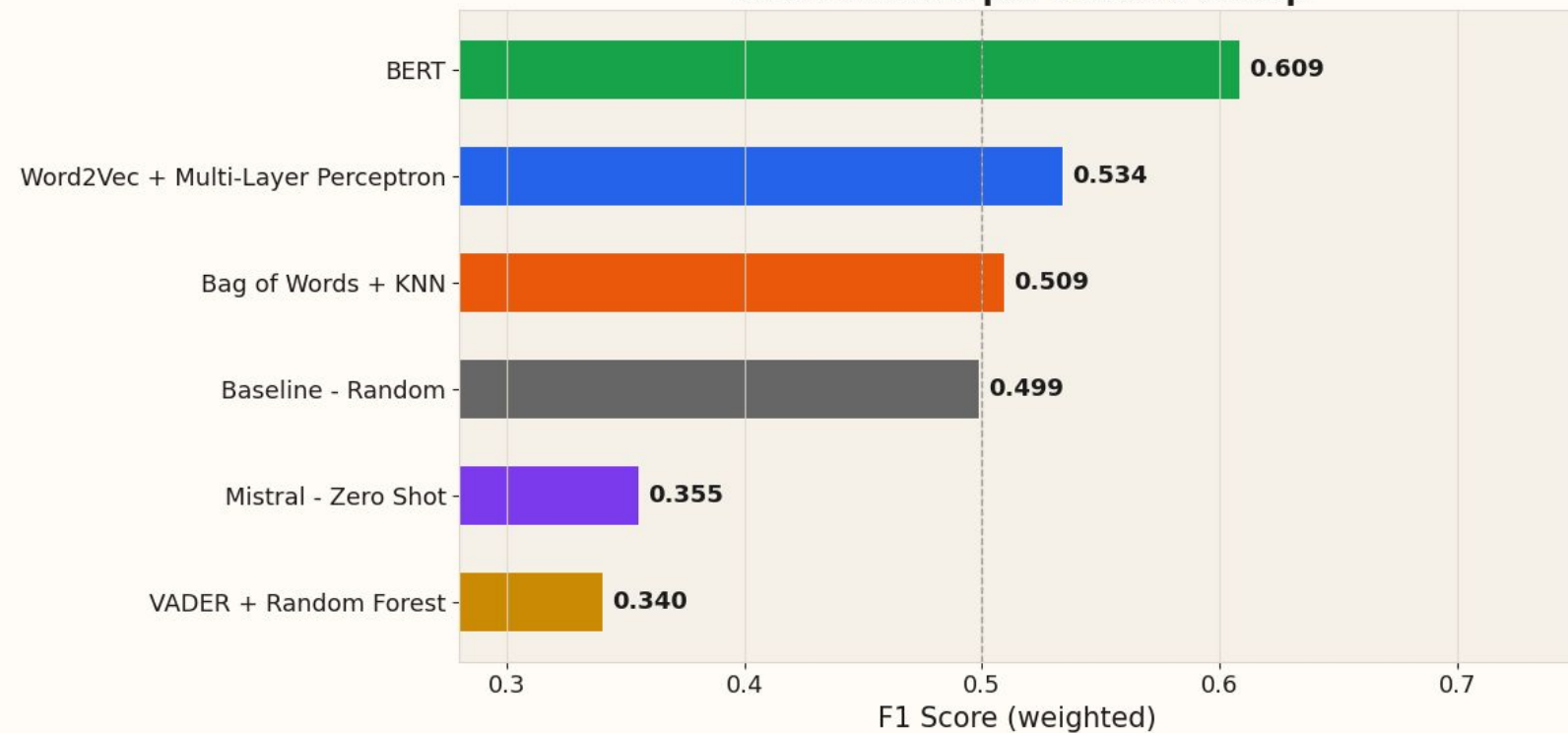


Results: Top 2 Models' Yearly Accuracy



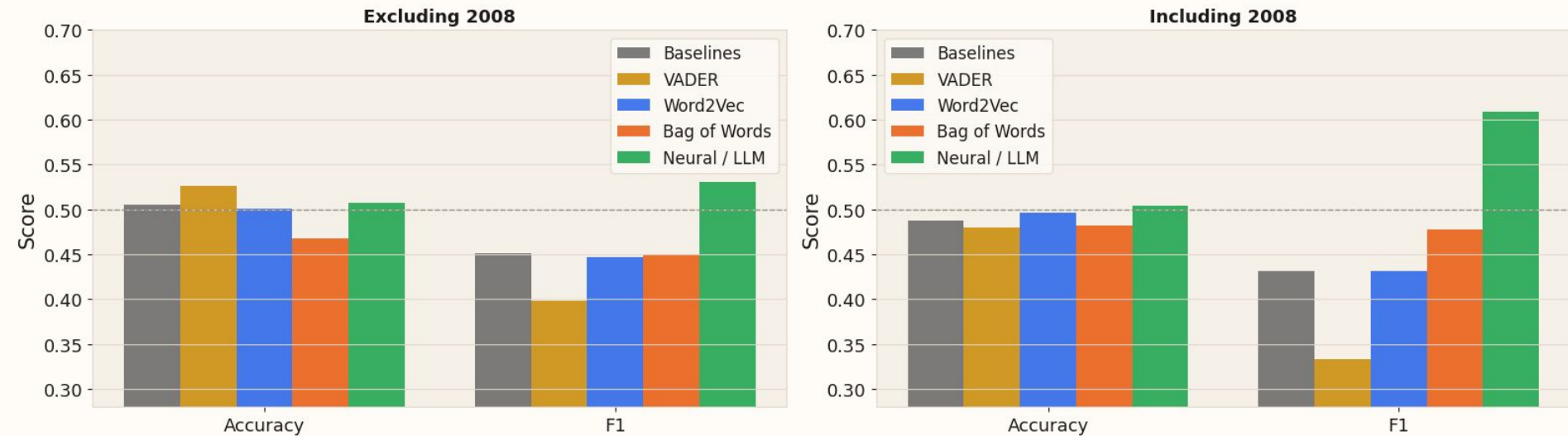
Results: F1

Best F1 Score per Feature Group



Results: Negligible Impact of 2008

Average Metrics by Feature Group



Results: Summary

Group	Model	Accuracy	F1
Bag of Words	Bag of Words + KNN	0.509	0.509
Bag of Words	Bag of Words + CNN	0.494	0.486
Bag of Words	Bag of Words + Logistic Regression	0.478	0.473
Bag of Words	Bag of Words + Multilayer Perceptron	0.470	0.463
Bag of Words	Bag of Words + Naive Bayes	0.460	0.456
Baselines	Baseline - Random	0.499	0.499
Baselines	Baseline - Momentum	0.481	0.480
Baselines	Baseline - Majority	0.483	0.315
Neural / LLM	BERT	0.504	0.609
Other	Mistral - Zero Shot	0.514	0.355
Other	Mistral - One Shot	0.514	0.351
Other	Mistral - Three Shot	0.456	0.286
VADER	VADER + Random Forest	0.476	0.340
VADER	VADER + Logistic Regression	0.478	0.338
VADER	VADER + CNN	0.486	0.321
Word2Vec	Word2Vec + Multi-Layer Perceptron	0.550	0.534
Word2Vec	Word2Vec + KNN	0.499	0.491
Word2Vec	Word2Vec + Linear SVM	0.499	0.452
Word2Vec	Word2Vec + Naive Bayes	0.455	0.419
Word2Vec	Word2Vec + Logistic Regression	0.491	0.381
Word2Vec	Word2Vec + CNN	0.483	0.315

Note: Mistral results are very similar is across all shots because it guessed 0 for almost every label!

*Including 2008 Data

Conclusions:

- Predicting DJIA direction from headlines alone is difficult
- Most models performed near baseline
- BERT offered the best results with a comparable accuracy and a significantly stronger F1
- Bag of Words, BERT, and Word2Vec were among the stronger learned feature approaches
- Removing 2008 had minimal impact on overall performance
- Stronger prediction likely needs richer features and more data

Key Takeaway: Headline text showed limited standalone predictive power for daily market movement

Future Work:

Different News Sources:

- Use headlines from sources other than Reddit
- Combine headlines across multiple sources

Update with current data:

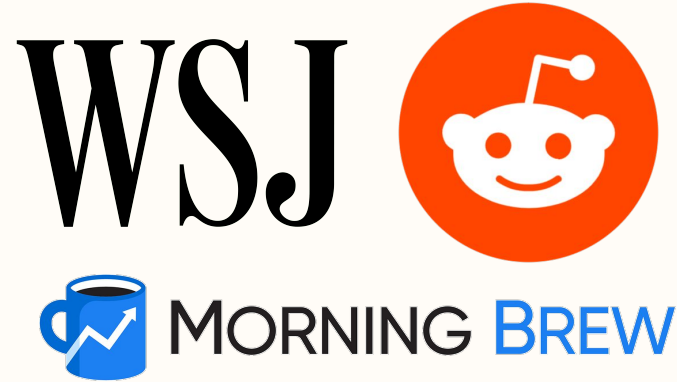
- Test on current and upcoming days and evaluate performance

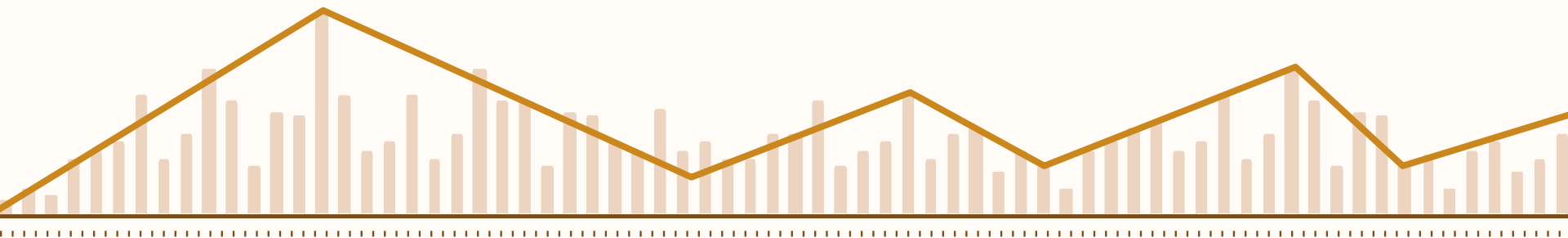
Experiment with more sentiment models:

- Experiment with sentiment models other than VADER which may perform better (like FinBERT)

Experiment with different data split:

- Split testing/training data chronologically rather than randomly





Q&A